

Q:7(histogram equalization, transformation function, uniform PDF)

#now

7. Define histogram equalization and explain its effects. Show that transformation of intensity histogram results in a uniform PDF. The intensity values of an image have the PDF as:

Equation of $p(r)$

Find the transformation function for equalizing the image histogram.

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Define histogram equalization and explain its effects.

Defination:

Histogram Equalization is a digital image processing technique that evenly distributes the pixel intensities throughout the entire range (0 to 255). This technique increases the image contrast and also improve the image details.

Histogram equalization Its Effect:

Contrast : maximizes contrast by stretching the image's **dynamic range** to occupy the full 0–255 intensity scale.

Details : enhances **feature visibility** by pulling out hidden textures from extremely dark or bright regions.

Histogram : redistributes pixel frequencies to create a **Uniform PDF (flat histogram)** where all intensities are equally likely.

Noise : can cause **over-enhancement** and unwanted **noise amplification** in smooth or low-signal areas.

Show that transformation of intensity histogram results in a uniform PDF

Goal: Show that $s = T(r)$ produces $p_s(s) = 1$.

1. **Define Transformation:** $s = T(r) = \int_0^r p_r(w)dw$

2. **PDF Relationship:** $p_s(s) = p_r(r) \left| \frac{dr}{ds} \right|$

3. **Differentiate s :** $\frac{ds}{dr} = \frac{d}{dr} \int_0^r p_r(w)dw = p_r(r)$

4. **Substitute:** $p_s(s) = p_r(r) \cdot \frac{1}{p_r(r)} = 1$

Conclusion: Since $p_s(s) = 1$, the output PDF is **uniform**.

transformation function:

Given: $p_r(r) = -2r + 2$ for $0 \leq r \leq 1$.

1. **Formula:** $s = \int_0^r p_r(w)dw$

2. **Integrate:**

$$s = \int_0^r (-2w + 2)dw = [-w^2 + 2w]_0^r$$

3. **Result:**

$$s = (-r^2 + 2r) - (0) = -r^2 + 2r$$

Final Answer: $s = -r^2 + 2r$ for $0 \leq r \leq 1$.